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Precise Measurements of Thermophysical Properties of Liquid Ti–6Al–4V (Ti64) Alloy On Board the International Space Station

Markus Mohr, Rainer Wunderlich, Rada Novakovic, Enrica Ricci, and Hans-Jörg Fecht*

The titanium alloy Ti–6Al–4V (Ti64) is a very widely applied alloy with extreme importance in the fields of aerospace, automotive, and biomedicine. For a large range of manufacturing methods that involve the liquid phase, such as casting, metal injection molding, thermal spraying, and additive manufacturing, process simulations are an important tool for the optimization and development of the fabrication process. Reliable thermophysical property data of Ti64 in the solid and the liquid phase are necessary inputs to such models. Due to the high melt reactivity of titanium alloys with container materials, measurements performed using conventional thermophysical equipment are usually impossible. Precise benchmark measurement of thermophysical properties (surface tension, viscosity, specific heat, total hemispherical emissivity, thermal conductivity, electrical resistivity, mass density) of Ti64 is presented in the undercooled liquid phase using containerless electromagnetic levitation on board the International Space Station (ISS). The obtained values are compared with predictive models that can help to analyze the data for other compositions of the alloy system. The property values are applied to simulations of industrial fabrication processes, to optimize these processes and thus the product quality.

1. Introduction

Titanium alloys generally exhibit high strength and toughness within a large composition and temperature range. Combined with their high corrosion resistance and low density, titanium alloys are important for many applications, ranging from lightweight constructions over aerospace applications to biomedical implants.[1–5] One of the most commonly used titanium alloys is the alpha–beta titanium alloy Ti64 (Ti–6Al–4V, in wt% or Ti–10.2Al–3.6V, in at%). Developed in 1954, it is an established alloy and a prototype for a large group of newer alpha–beta Ti–Al alloys with good strength to weight ratio.[2,5,6] Different fabrication routes to produce parts and components from Ti64 exist (casting, additive manufacturing, metal injection molding, etc.), mainly including the presence of the liquid phase.[6] Ti64 is also very commonly used in 3D printing and additive manufacturing technologies that continue to evolve rapidly.

The final microstructure, defect density, residual stresses, surface roughness, porosity, and consequently the mechanical properties of Ti64 parts are depending on the used process conditions.[7–12] Manufacturing processes can be optimized by simulations of heat and mass flow during the manufacturing process.[13–17] Also, the microstructure evolution, including porosity and other defects, can be predicted by numerical methods.[18] These simulations need precise thermophysical property data of the simulated alloy as input parameters.[5,13–18] Although Ti64 is a widely applied alloy, not too much literature data are available for its thermophysical properties in the liquid phase. Also, there are some disagreements in the published literature concerning the thermophysical property data, such as thermal conductivity or specific heat capacity. One reason is that the measurement of thermophysical properties of liquid Ti alloys using classical thermoanalytical equipment (container-based methods) is challenging or even impossible. The inevitable contact of the material with the container walls leads to contaminations and parasitic container-wall reactions that will prevent precise measurements of thermophysical properties.[19]

We present measurements of Ti64 using the electromagnetic levitator (ISS-EML) inside the European science module "Columbus" on board the International Space Station (ISS). This containerless method is the methodology of choice for precise measurement of thermophysical properties of high-temperature reactive liquid metallic melts. The microgravity environment is used to diminish the strong forces necessary

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to lift and position the sample under 1 g conditions. This also enables measurement conditions with a spherical sample exhibiting only laminar flows inside the droplet. The microgravity environment allows the decoupling of the sample temperature from the positioning forces which enables a good temperature control and high undercooling of the sample.

Our aim is to present benchmark measurements of thermophysical properties, here of Ti64 (sample #12, Batch-1). The thermophysical property data can not only be used to model the processing of the measured alloy composition, but is also a contribution to the validation of models that can predict property values of other Ti–Al compositions.

Therefore, in this work, the surface tension of liquid Ti64 is compared with model predictions obtained by geometrical models as well as by the Compound Formation Model (CFM), the Butler model,[19] and Quasi Chemical Approximation (QCA) for regular solution.[20] Further comparison of Ti64 data with respect to those of the main binary subsystem, i.e., the Al–Ti, is also done.[21,22] Similarly, the new experimental viscosity data of liquid Ti64 alloy are compared with the viscosity isotherm calculated by the Moelwyn–Hughes (MH) model.[23]

Several measurements of thermophysical properties of Ti64 are reported in the literature, most of them using unconventional measurement equipment, such as electromagnetic and electrostatic levitation[21,24] or subsecond pulse calorimetry.[25–27]

Kaschnitz et al.[25] applied a millisecond and microsecond measurement method and obtained the specific heat capacity of Ti64 in the solid and liquid phase, as well as the electrical resistivity. Boivineau et al.[26] also used a pulse heating system to obtain thermophysical properties of Ti64 in the solid and liquid state. Milošević and Aleksic[27] investigated the Ti64 alloy by a pulse heating system in the solid phase. Li performed calorimetric measurements using an electrostatic levitator, which allowed to obtain the ratio between specific heat capacity and total hemispheric emissivity c_p/ε for Ti64 in the liquid and undercooled liquid phase.[24] Density and surface tension of liquid Ti64 was obtained by electromagnetic levitation by Egry et al. on ground and under microgravity conditions.[21] The electromagnetic levitation technique was already successfully applied on other engineering alloys, such as on Ti–Al alloys (Wunderlich et al.[28]) during parabolic flights (PF), on Ni-based superalloys during PF (Higuchi et al.[29]), and on board the ISS (Mohr et al.[30]).

2. Experimental Section

The electromagnetic levitation facility ISS-EML was already described elsewhere,[31–33] therefore only the most important aspects are briefly described here. The ISS-EML is equipped with an exchangeable sample chamber and an experiment chamber. The sample chamber contains 18 different samples, which are spheres of a diameter of $d = 6$–8 mm. The experiment chamber contains the coil system for positioning and heating (a SUPOS coil system[32]). A high-speed camera from top (the axial direction) and one from the side (the radial direction) were observing the levitating sample with typical frame rates of 150 and 400 Hz, respectively. A pyrometer is integrated into the axial camera, such that the sample temperature can be measured at one sample pole, whereas the coils arrangement leads predominantly to a heating at the sample equatorial region.

We performed measurements of surface tension, viscosity, mass density, specific heat capacity, total hemispheric emissivity, and electrical resistivity of Ti–6Al–4V as a function of temperature in the stable liquid phase and in the undercooled liquid phase (≈200 K undercooling).

A typical measurement cycle started with the heating, melting, and overheating of the sample. Subsequently, the heating power was reduced and the sample was able to cool while freely levitating inside the gas atmosphere (here Argon, 3.5×10^4 Pa). During this cooling phase, several methods were applied to obtain the aforementioned physical quantities.

2.1. Electrical Resistivity and Mass Density

The electrical resistivity was determined applying the measurement method developed by Lohöfer.[34,35] The impedance of the oscillating circuit consists of the heater coil and a parallel capacitor. The impedance of the coil is altered by the presence of a sample and the oscillator impedance is measured during the cooling of the sample by a so-called sample coupling electronics. From a measurement without sample, the circuit parameters can be determined and consequently the impedance of the sample can be obtained.[35] The sample impedance Z_S is related to the heater oscillator frequency ω_{Htr} resistivity ρ_e and radius a of the sample as[35]

$$Z_S(\omega_{Htr}, a, \rho_e) = C_c \cdot \omega_{Htr} \cdot a^3 \left[\frac{1}{q} - \frac{1}{q^2} + i\left(\frac{1}{q} - \frac{2}{3}\right)\right] \tag{1}$$

with

$$q(\omega_{Htr}, a, \rho_e) = a\sqrt{\frac{\mu_0 \cdot \omega_{Htr}}{2\rho}} \tag{2}$$

where μ_0 is the magnetic permeability of free space. After some rearrangement of Equation (1) and (2), the specific resistivity ρ_e and radius a of the sample can be obtained as a function of sample temperature T.[35] Necessary is, however, the knowledge of the coil constant C_c. This was obtained by calibration with a Zr sample of known resistivity and diameter. In addition, the calibration was checked by extrapolation of the temperature-dependent sample radius down to the room temperature, where the sample radius has been measured on ground.

2.2. Surface Tension and Viscosity

Surface tension γ and viscosity η were determined by applying the oscillating drop method.[36,37] Short heater pulses were applied while the sample cooled down, which initiated surface oscillations. An edge-detection algorithm on the recorded high-speed videos revealed the samples circumference and therefore the surface oscillation amplitude $\delta(t)$. Fourier transformations applied in several time frames during the sample cooling allowed the determination of the surface oscillation frequency f_{osc}. The relative amplitudes $\delta(t)/d$ of the surface oscillations were small enough to prevent nonlinear effects.[38] Therefore, the frequency f_{osc} of the surface oscillations can be used to obtain the surface tension γ of the sample by

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$$\gamma = \frac{3}{8}\pi M f_{osc}^2 \tag{3}$$

where M is the mass of the sample.[39] The internal friction in the liquid sample led to a damping of the surface oscillations. The time constant τ of the exponential decay of the surface oscillation amplitude can be used to obtain the samples viscosity using Lambs equation[40]

$$\eta = \frac{3}{20\pi}\frac{M}{R}\frac{1}{\tau} \tag{4}$$

where R is the average sample radius.

2.3. Specific Heat Capacity and Total Hemispherical Emissivity

The samples heat capacity C_p can be obtained by the modulation calorimetry method.[41–44] For the case of a spherical sample in an electromagnetic levitator, a heater power modulated with frequency f_{mod}, that is having the form

$$P(t) = P_{av} + \Delta P \cdot \sin(2\pi f_{mod} t + \varphi_0) \tag{5}$$

will lead to a temperature signal of the form

$$T(t) = T_{av} + \Delta T \cdot \sin(2\pi f_{mod} t + \varphi_1) \tag{6}$$

In addition, a stepwise change of average power will lead to an additional exponential increase or decrease in the average temperature $T_{av}(t) = \Delta T_{av} \cdot e^{-\frac{t}{\tau_1}}$. The relation between the temperature amplitude ΔT and the power modulation amplitude ΔP is simply[41]

$$C_p = \frac{\Delta P}{2\pi f_{mod}\Delta T} \cdot f_c(\tau_1, \tau_2, f_{mod}) \tag{7}$$

with the heat capacity C_p of the sample and a correction function $f_c(\tau_1, \tau_2, f_{mod})$, which depends on the external relaxation time τ_1 and the internal relaxation time τ_2[41]

$$f_c(\tau_1, \tau_2, f_{mod}) = (1 + (2\pi f_{mod}\tau_1)^{-2} + (2\pi f_{mod}\tau_2)^2)^{-1/2} \tag{8}$$

For the present sample, the internal relaxation time (a time constant characteristic for the heat flow from the directly heated equatorial region to the poles) is typically in the range of 0.1 s, whereas the external relaxation time depends on the sample surface, emissivity, and the used gas atmosphere and it is in the order of several seconds.

The external relaxation time can be obtained from the temperature response to a step change of average power consumption. **Figure 1** shows a typical modulation calorimetry cycle, where on the left ordinate the temperature is plotted against time. On the right ordinate, the coil current square is shown as a function of time. It can be seen that all modulations are performed in the undercooled liquid phase.

To calculate the amplitude of power modulation ΔP, an analytical model for the power consumption of a spherical sample inside a symmetrical coil geometry was used.[45] The necessary input parameters for the model are the sample radius $a_0(T)$ and the sample specific resistivity $\rho_e(T)$ (both temperature dependent) and the frequency ω_{Htr} of the heater radio frequency (RF) field. Then, a coupling coefficient can be defined as

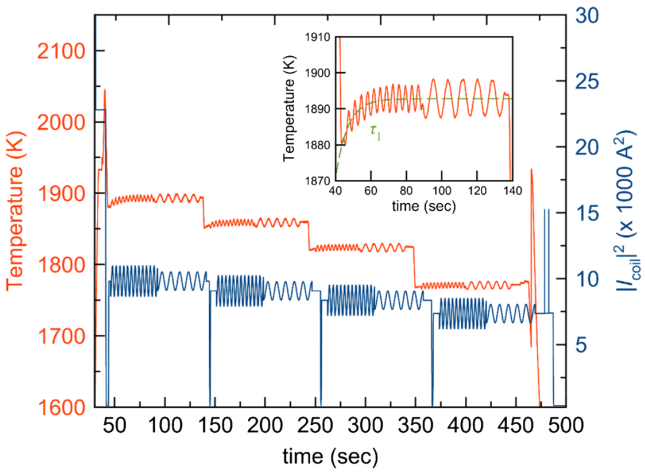

Figure 1. One typical modulation calorimetry cycle performed on Ti64. The left ordinate shows the temperature readings as a function of time (red), whereas on the right ordinate the square of the coil current is shown (blue). The inset shows a detail of the transient temperature signal together with an exponential fit, used to obtain the external relaxation time τ_1.

$$G_H(\rho(T), a_0(T), \omega_{Htr}) = P/I_{coil}^2 \tag{9}$$

Further technical details can be found in the literature.[30,46] The method was successfully verified on a pure Zirconium sample, where specific resistivity and specific heat capacity are well documented in the literature.

The heat balance for a sphere suspended in an electromagnetic heating field can be described as

$$C_p \frac{\partial T}{\partial t} = P_0 - A\varepsilon\sigma T_0^4 - P_{gas}(T) \tag{10}$$

where C_p is the heat capacity of the sphere, P_0 is the electrical heating power, A is the surface area, ε is the total hemispherical emissivity, σ is the Stephan–Boltzmann constant, and $P_{gas}(T)$ labels the heat loss to a gas atmosphere by heat conduction. For the case of vacuum $P_{gas}(T) = 0$ and for the limit of small temperature changes, the problem can be linearized using a Taylor expansion to obtain the time-dependent temperature change after a change in average heating power.[41] The resulting time-dependent temperature can be given as $T(t) = T_0 + \Delta T(t)$, with $\Delta T(t) = \exp(-t/\tau_1)$. In the vacuum case, the external relaxation time τ_1 can be described as

$$\tau_1 = \frac{C_p}{4A\varepsilon\sigma T_0^3} \tag{11}$$

For this investigation, the measurement cycles were performed in Argon atmosphere. This way, the heat loss model by Lohöfer and Schneider[47] is used to describe $P_{gas}(T)$. Consequently, this leads to the expression

$$\tau_1 = \frac{C_p}{16\pi R^2\varepsilon\sigma T_0^3 + \frac{4\pi R}{1-\frac{R}{b_g}}\lambda_{0,g}T_0^{\alpha_g}} \tag{12}$$

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where R is the radius of the sample, and b_g, $\lambda_{0,g}$, and α_g are parameters describing the thermal conductivity of the gas. With known parameters of b_g, $\lambda_{0,g}$, and α_g for Argon,[47] the total hemispherical emissivity can also be calculated from the external relaxation times measured in the process cycles performed in Argon.

3. Thermodynamic Modelling

3.1. Surface Tension

Preliminary thermodynamic analysis of Al–Ti, Al–V, Ti–V, and Al–Ti–V liquid alloys together with the calculations of their surface properties indicates that the surface tension of Ti64 can be accurately predicted by the geometric and Butler's models.[21] The Kohler, Toop, and Chou geometric models and their mathematical formalism were detailed in the literature,[48] and thus only the basic equations are given in the current work. To calculate the surface tension of ternary alloys, the key property of the aforementioned models is the excess surface tension, ${}^{xs}\gamma$, obtained from the corresponding property of the binary subsystems, expressed by Equation (13)–(15), respectively, as

$$\begin{aligned}{}^{xs}\gamma = {} & (X_1+X_2)^2({}^{xs}\gamma_{12})_{X1/X2} + (X_1+X_3)^2({}^{xs}\gamma_{13})_{X1/X3} \\ & + (X_2+X_3)^2({}^{xs}\gamma_{23})_{X2/X3}\end{aligned} \tag{13}$$

$$\begin{aligned}{}^{xs}\gamma = {} & \frac{X_2}{1-X_1}\cdot{}^{xs}\gamma_{12}(X_1;1-X_1) + \frac{X_3}{1-X_1}\cdot{}^{xs}\gamma_{13}(X_1;1-X_1) \\ & + (X_2+X_3)^2\cdot{}^{xs}\gamma_{23}\left(\frac{X_2}{X_2+X_3};\frac{X_3}{X_2+X_3}\right)\end{aligned} \tag{14}$$

$${}^{xs}\gamma = \frac{X_1X_2}{X_{1(12)}X_{2(12)}}\cdot{}^{xs}\gamma_{12} + \frac{X_2X_3}{X_{2(23)}X_{3(23)}}\cdot{}^{xs}\gamma_{23} + \frac{X_3X_1}{X_{3(31)}X_{1(31)}}\cdot{}^{xs}\gamma_{31} \tag{15}$$

where X_i and x_i represent the molar fraction i in the ternary and binary systems, respectively.

Similarly, the surface tension of ternary alloys can be calculated by the Butler model combining the surface tension of binaries[19,21] as follows

$$\begin{aligned}\gamma = {} & \gamma_i + \frac{RT}{S_i}\ln\frac{X_i^s}{X_i^b} \\ & + \frac{1}{S_i}\left[G_i^{xs,s}\left(T, X^s_{j(j=2,3)}\right) - G_i^{xs,b}\left(T, X^b_{j(j=2,3)}\right)\right],\ i=1,2,3\end{aligned} \tag{16}$$

where $G_i^{xs,s}(T, X^s_{j(j=2,3)})$ and $G_i^{xs,b}(T, X^b_{j(j=2,3)})$ are temperature-dependent partial excess Gibbs energies of component i of the surface and bulk phases, respectively, and $X^s_{j(j=2,3)}$ and $X^b_{j(j=2,3)}$ are corresponding compositions. T, R, γ_i, and S_i are temperature, gas constant, surface tension, and surface area of component i, respectively.

3.2. Viscosity

There are many empirical and semiempirical viscosity models[23] and after preliminary calculations, the MH model has shown to be the most appropriate to predict the viscosity of liquid Al–Ti alloys. The MH viscosity isotherm is described by

$$\eta = (x_1\eta_1 + x_2\eta_2)\left(1 - 2x_1x_2 - \frac{\Omega}{RT}\right) \tag{17}$$

where η_1 and η_2 are the viscosities of pure components and Ω is the mixing enthalpy parameter of the alloys.

4. Results

Figure 2a shows the electrical resistivity (ρ_e) as a function of sample temperature in the solid, as well as in the liquid and undercooled liquid phase. The data points (red circles) were smoothed by 10 points averaging. The scattering of the original data is owed to the low sample viscosity (see **Figure 3**b) and the resulting strong sample oscillations in the stable liquid phase. The specific electrical resistivity can be described as

$$\rho_{e,l}(T) = \{(180.8 \pm 2.2) – (1.12 \pm 0.12) \times 10^{-2} \times T[K]\}\ \mu\Omega\,\text{cm} \tag{18}$$

in the stable and undercooled liquid phase (1750–2000 K). The specific resistivity at the liquidus temperature is $\rho_{e,l}(T_{liq}) = (158 \pm 2)\ \mu\Omega\,\text{cm}$ in the liquid phase and $\rho_{e,s}(T_{liq}) = (162 \pm 1)\ \mu\Omega\,\text{cm}$ in the solid phase. In the solid phase, for temperatures ranging from 1650 to 1900 K, the electrical resistivity can be described by

$$\rho_{e,s}(T) = \{(152.7 \pm 0.4) + (4.76 \pm 0.23) \times 10^{-3} \times T[K]\}\ \mu\Omega\,\text{cm} \tag{19}$$

The electrical resistivity is abruptly increasing during sample cooling due to the β→α phase transformation at around 1100 K. The error bars given in Equation (18) and (19) describe the confidence interval for a confidence level of 95%. A systematic influence on the measurement accuracy of electrical resistivity and density measurement is the geometry factor C_c, which is calibrated by the measurement of a solid Zr sample (reference sample). Random measurement errors, leading to the small scatter of data, origin mainly from small sample oscillations and relative movements. The measurement uncertainty of the electrical resistivity values is on the order of 0.05%.[35] However, the data scatter due to sample movements and oscillations is dominating the measurement uncertainty, making the given confidence intervals good measures of the measurement uncertainty.

Using the known sample mass and the sample diameter, the mass density was calculated (Figure 2b). The smoothed data (10 points, moving average) show a linear increase of mass density with decreasing temperature before and after the $\beta \rightarrow \alpha$ phase transition. The mass density in the solid phase $\rho_{m,s}$ in the temperature range of 1650–1900 K can be expressed as

$$\begin{aligned}\rho_{m,s} = {} & \{(4.163 \pm 0.003) – (0.145 \pm 0.001)\,10^{-3} \\ & \times (T - T_{liq})\}\ \text{g cm}^{-3}\end{aligned} \tag{20}$$

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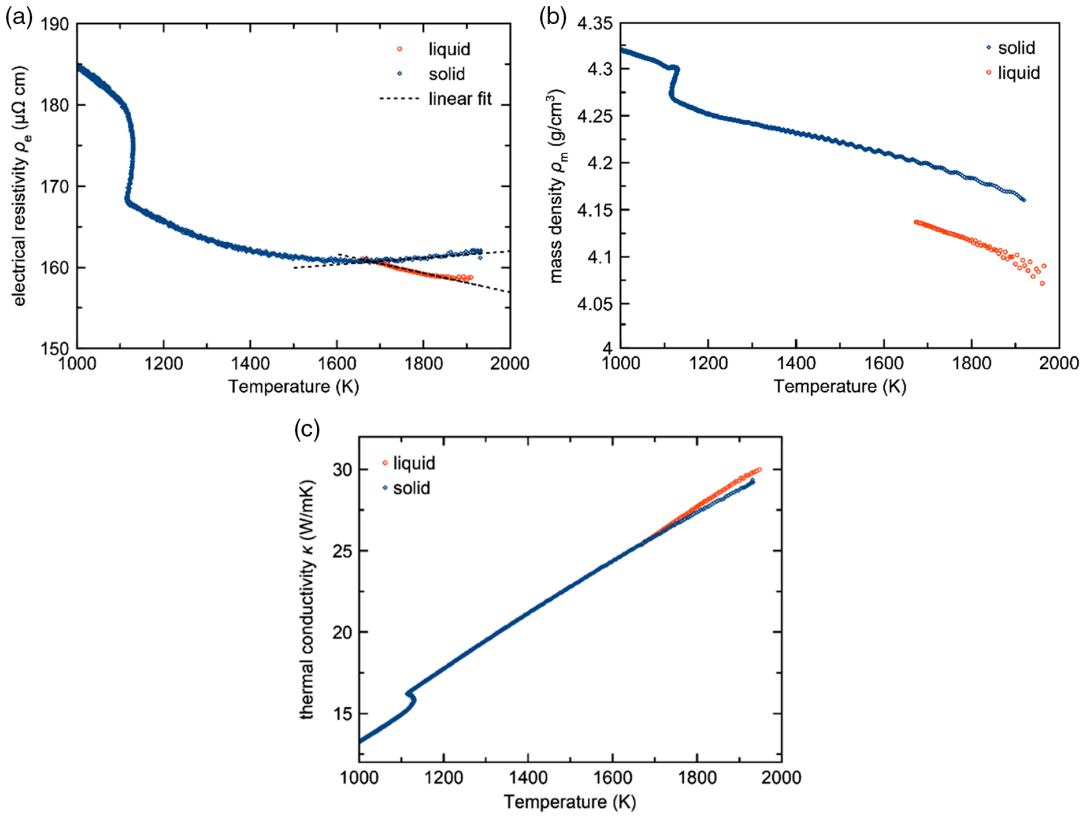

Figure 2. a) Specific electrical resistivity in the liquid and solid phase b) mass density of Ti64 in the stable and undercooled liquid and solid phase c) thermal conductivity of Ti64 in the solid and liquid phase.

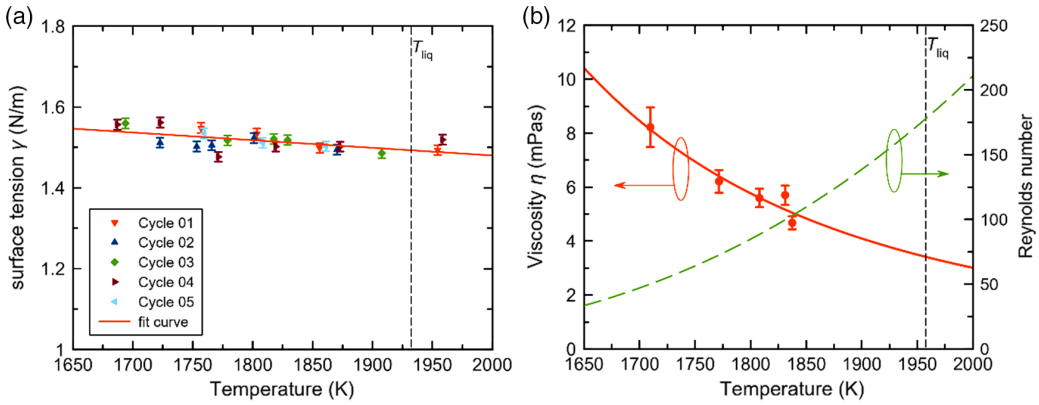

Figure 3. a) Surface tension of Ti64 in the stable and undercooled liquid phase. b) Left ordinate: viscosity of Ti64 in the undercooled liquid phase, right ordinate: estimated Reynolds number in the droplet as a function of temperature (dashed line).

The mass density in the liquid phase $\rho_{m,l}$ can be fitted by

$$\rho_{m,l} = \{(4.087 \pm 0.001) - (0.204 \pm 0.007)10^{-3} \times (T - T_{liq})\}\,\mathrm{g\,cm^{-3}} \tag{21}$$

The uncertainties stated in Equation (20) and (21) are denoting the confidence interval of the fitting, for a confidence level of 95%. The measurement uncertainty of the density measurement mainly origins from the residual sample deformation during the measurement. The sample elongation was measured from video recordings to be of only about 0.3%, which corresponds to a measurement uncertainty of the density of about 0.2%.

The Wiedemann–Franz law

$$\kappa \times \rho_e = L \times T \tag{22}$$

relates the electrical resistivity ρ_e to the thermal conductivity κ, where L is the theoretical Lorentz number,

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$L = 2.45 \times 10^{-8}\,\mathrm{W\,\Omega\,K^{-2}}$. Under the assumption, that the heat conduction is mainly governed by electrons, the thermal conductivity of the sample can be approximated from the measured electrical resistivity. The result is shown in Figure 2c.

The temperature-dependent surface tension obtained in five processing cycles is shown in Figure 3a. The temperature dependence of the surface tension can be expressed as

$$\gamma(T) = \{(1.493 \pm 0.008) - (1.9 \pm 0.5)10^{-4} \times (T - 1933\,K)\}\,\mathrm{N\,m^{-1}} \tag{23}$$

in the stable and undercooled liquid phase, in a temperature range of 1700–1950 K.

The viscosity values obtained from the damping time constants determined in the undercooled liquid phase are shown in Figure 3b. Due to the relatively low viscosity, varying between about 5 and 8 mPa s in the temperature range from 1700 to 1850 K, the surface oscillations were only weakly damped directly after the heater was turned off and as such, the damping time constant could only be determined in the undercooled liquid phase. Using an Arrhenius fit in the investigated temperature range, the viscosity at T_{liq} can be approximated to be about 4 mPa s.

The measurement uncertainty of the surface tension is determined by the accuracy of the surface oscillation frequency determination and the accuracy of the sample mass measurement. The sample mass of 0.638 g was measured with an uncertainty of ±0.005 g. The oscillation frequency can be determined with an uncertainty of ±0.05 Hz. These measurement uncertainties are used to calculate the error bars in Figure 3a. In addition, sample rotations can lead to a small shift of the surface tension,[49] this could explain the small scatter of the measurement data.

The uncertainty of the viscosity measurement is determined by the accuracies of the sample mass, radius, and the damping time constant. The sample mass is measured with uncertainty of ±0.005 g; the sample radius was measured with an uncertainty of ±0.03 mm. The uncertainty of the damping time constant is assumed to be about $\Delta\tau = \pm 0.1$ s. The resulting error bars are shown in Figure 3b. In addition, the absence of laminar flow conditions could lead to deviations from the real value. To verify the laminar flow conditions, we used a model[50] to estimate the Reynolds number of the flow in the droplet during the measurement. As shown in Figure 3b, the Reynolds number is always far below the value of 600 for which a transition to turbulent flow is expected.[51]

The results of the modulation calorimetry cycles are sown in **Figure 4**a, showing molar heat capacity values measured in the solid phase and in the undercooled liquid phase. The uncertainties origin from the uncertainties of the temperature oscillation amplitude (±0.1 K), as well as on the uncertainty of the sample mass (±0.005 g). The calculated amplitude of power dissipation is compared with measurements on a solid Zr sample, thus the power dissipation is calibrated. From this comparison, a remaining uncertainty of power dissipation is determined to be about 2%. Therefore, the uncertainty of the measured molar heat capacity is about 2.4%.

The external relaxation time τ_1 was used to obtain the total hemispherical emissivity. The result for the total hemispheric emissivity in the solid and liquid phase is shown in Figure 4b. The measurement uncertainty is here independent of the mass, and therefore only determined by the uncertainty of the sample radius, as well as the uncertainty of the measured specific heat. In addition, the accuracy of the applied model extension was tested by the measurement of ε on a solid Zr reference sample in vacuum and in Argon atmosphere. The results deviated less than 1.4%. On this basis, the measurement uncertainty of the total hemispherical emissivity was determined (see error bars in Figure 4b).

5. Discussion

The electrical resistivity we obtained in the liquid phase is in very good agreement with the values obtained by Boivineau et al.[26] The specific electrical resistivity in the solid phase is lower than the values by Milošević and Aleksic[27] and lower than the values by Kaschnitz et al.[25] However, the resistivity in the solid state is dependent on the actual microstructure, which is a function of thermal history of the sample and can therefore differ between the different literature reports.

The obtained solid and liquid density is in good agreement with the values obtained by Li et al.[52] using an optical method

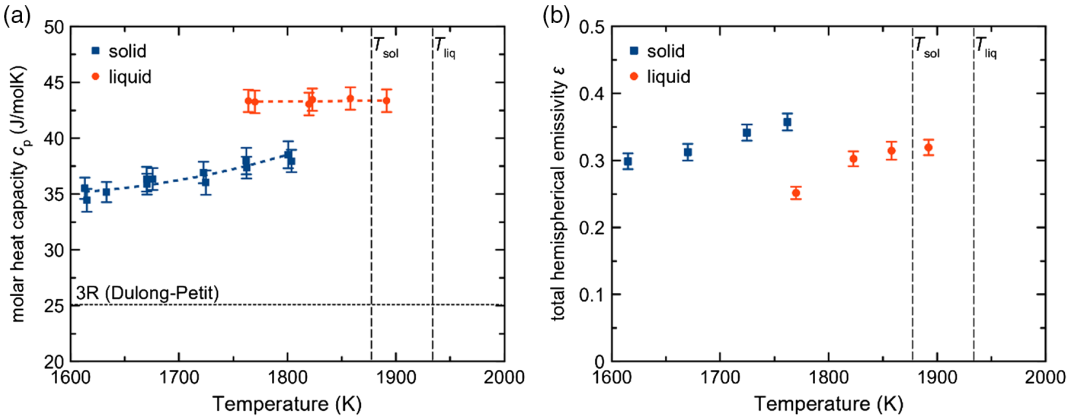

Figure 4. a) Molar heat capacity of Ti64 in the solid phase and undercooled liquid phase b) total hemispheric emissivity of Ti64 as a function of temperature.

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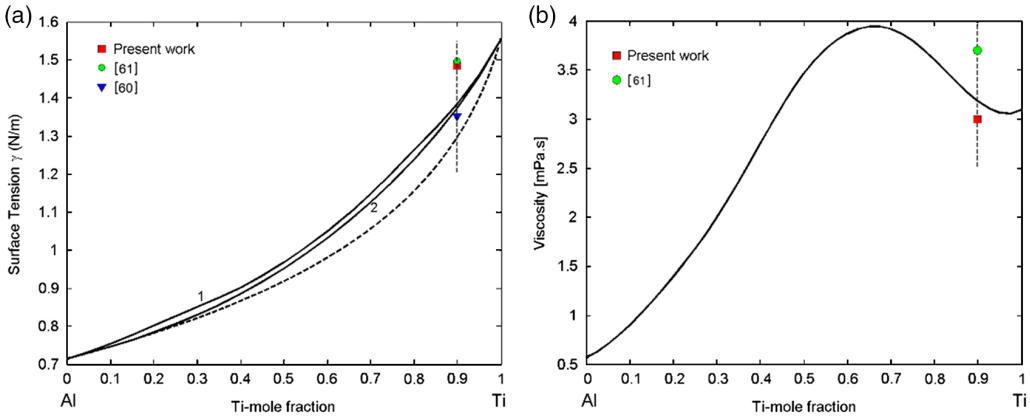

Figure 5. a) Experimental surface tension data of liquid Ti64 alloy obtained in the present work by EML in ISS and literature data measured by ground-based EML[60] and in PF,[61] respectively. The Al–Ti surface tension isotherms (1-CFM; 2-Butler/QCA for regular solution, the ideal solution model (dashed line)) calculated at $T = 1973$ K are shown for a comparison. (b) Experimental viscosity data of liquid Ti64 alloy obtained in the present work by EML in ISS and literature data measured during PF.[61] The Al–Ti viscosity isotherm, calculated by the MH model for $T = 1973$ K is shown for a comparison.

during electrostatic levitation. The handbook values of the mass density in the solid phase[53] are in good agreement with our measurements, whereas the handbook[53] gives a 28–32% lower value for the liquid density than our measurement and the measurements of Li et al.[52]

We obtained the thermal conductivity of Ti64 in the solid, undercooled, and stable liquid phase. The thermal conductivities of the solid and stable liquid phase are in agreement with the values reported by Boivineau et al.,[26] who also calculated the thermal conductivity using the assumption of the Wiedemann–Franz law. However, our result of thermal conductivity in the liquid phase, obtained under the assumption that thermal transport is dominated by electron transport, is smaller ($\kappa = 29.8\ \mathrm{W\,m^{-1}\,K^{-1}}$, $T = 1933$ K) than reported data of thermal conductivity ($\kappa = 33.3\ \mathrm{W\,m^{-1}\,K^{-1}}$, $T = 1919$ K) that were obtained from thermal diffusivity measurements.[53]

The analysis of the Gibbs free energies of the Al–Ti,[54] Al–V,[55] and Ti–V[56] liquid phases indicates moderate interactions in the case of the first two systems and weak interactions between constituent atoms in the last one. The Al–Ti and Al–V phase diagrams show the presence of intermetallics in these systems, and therefore the most appropriate model to describe their surface tension would be the CFM,[19] whereas the Butler model describes accurately liquid Ti–V alloys having a weak demixing tendency.[21] Concerning the surface tension of liquid Al–Ti alloys,[22] the difference between the values predicted by the CFM and QCA model (in this case the QCA calculations are overlapping with those obtained by the Butler model) is about 1.1% (**Figure 5**a). To obtain accurate surface tension predictions, the surface tension reference data of liquid Al,[57] Ti,[58] and V,[59] all obtained by oscillating drop method in EML were used in the present calculations. The input data are listed in **Table 1**. To establish the effects of the third component, V, on the surface tension of Ti64 alloy, a comparison with the corresponding Ti–Al binary alloy having the same Al-content as Ti64 was done. The surface tensions obtained in our study are higher than the values obtained in ground-based electromagnetic levitation experiments using the oscillating drop method[60] but similar to the values obtained by Wunderlich et al. obtained in electromagnetic levitation during PF,[61] as shown in Figure 5a. Literature data[61] and the experimental value obtained in this study of Ti64 (Ti–6Al–4V, in wt%; Ti–10.2Al–3.6V, in at%) are higher than those of the model predicted values for the Ti–10.2Al, in at% (Ti–6Al, in wt%) because the surface tension of V is significantly higher than that of Ti. Subsequently, the new surface tension data of Ti64 alloy were directly compared with the surface tension values of ternary Ti–Al–V alloys calculated by geometric models (**Table 2**) as well as by the Butler model. The experimental surface tension data of this study exhibit the best agreement with the corresponding value obtained by the Butler model. Indeed, the predicted surface tension value (Table 2) is close to the iso-surface tension line showing $1.45\ \mathrm{N\,m^{-1}}$ (**Figure 6**).

Table 1. Thermodynamic data of the excess Gibbs free energies (Redlich–Kister polynomial with ν terms) on liquid Al–Ti, Al–V and Ti–V alloys and the surface tension reference data of Al, Ti, and V pure metals.

System	ν	${}^{xs}G_L^{A-B}$ [J mol^{-1}]	Reference
Al–Ti	0	$-118\,048 + 41.972\,T$	[54]
	1	$-23\,613 + 19.704\,T$	
	2	$34\,757 - 13.844\,T$	
Al–V	0	$-69\,140 + 9.63\,T$	[55]
	1	12 700	
Ti–V	0	1400	[56]
	1	−4100	

Surface tension				
Pure metal	Melting point [K]	$\gamma 0$ [N m^{-1}]	$d\gamma/dT$ [10^{-3} N m^{-1} K^{-1}]	
Al	934	0.866	0.146	[57]
Ti	1941	1.560	0.165	[58]
V	2183	1.935	0.270	[59]

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Table 2. Model predicted values and the surface tension experimental data of liquid Ti64 alloy (Ti–6Al–4V, in wt%; Ti–10.2Al–3.6V, in at%) obtained at $T = 1973$ K.

Models				Experimental	
Kohler's	Toop's	Chou's	Butler's	Method	Data [N m^{-1}]
1.461	1.322	1.380	1.479	EML/ISS, present work	1.490 at $T = 1950$ K
					1.485[a)] calc. for $T = 1973$ K
				EML[60]	1.353
				PF[61]	1.497

[a)]Extrapolated value.

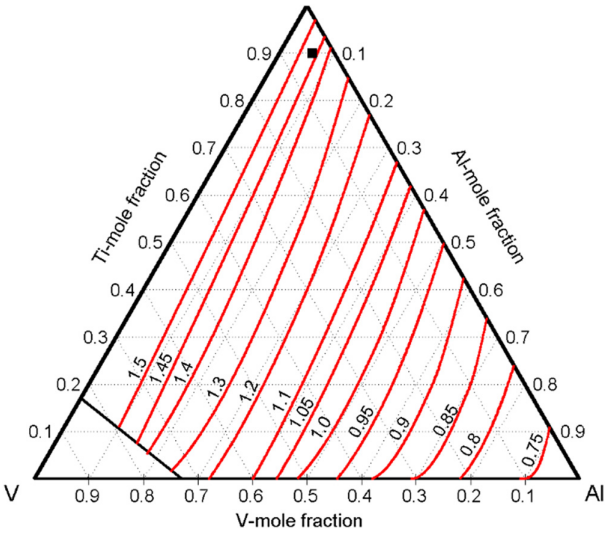

Figure 6. Iso-surface tension lines of liquid Al–Ti–V alloys calculated by the Butler model at $T = 1973$ K. In the Gibbs triangle, the square symbol indicates the composition location of the Ti64 (at%) and the nearest iso-surface tension line shows its model predicted surface tension value.

The obtained viscosity value in this study is lower than those measured by Wunderlich et al. on samples with different oxygen concentrations (0.047 and 0.096 wt%);[61] therefore, we expect the oxygen concentration in our sample to be even lower. Applying the viscosity models reported in previous studies,[23,62] preliminary calculations of Al–Ti isotherms indicate the MH model as the most appropriate. Considering the dataset on the enthalpy of mixing of liquid Al–Ti alloys[63,64] measured at $T = 2000$ K, together with the viscosity reference data of Al[62] and Ti,[65] the viscosity isotherm was calculated. Similarly, as in the case of the Ti64 surface tension data, the new viscosity data as well as that reported in ref. [61] are compared with the corresponding binary value obtained for the alloy composition with the same Al content. The experimental data of viscosity differ by about 5% from the calculated value, whereas literature data[61] are higher with respect to both the theoretical and the experimental values obtained in this study (Figure 5b).

Even though the suggested models for surface tension and viscosity of the Ti–Al system have been successfully applied for the Ni–Al-based system,[66] to describe the surface tension of Ni-based superalloys, more experimental data are needed to verify the validity of the suggested model.

The literature value for the specific heat capacity of Ti64, given by Mills[53] of 0.83 J g^{-1} K^{-1} (38.79 J mol^{-1} K^{-1}), is by about 10% smaller than our measured specific heat capacity of (0.93 ± 0.02) J g^{-1} K^{-1} (or molar heat capacity of (43.34 ± 1.00) J mol^{-1} K^{-1}). The specific heat capacity obtained in our measurement is well within the uncertainties of the measurements reported by Kaschnitz et al.[25] However, we have extended the measurement range by about 200 K into the undercooled liquid phase, which is not possible with container-based or subsecond pulse-heating measurement systems.

Literature data of total hemispheric emissivity of Ti64 in the literature do sometimes show influences of an oxide layer on the solid Ti64.[27,67] Measurement data of hemispheric emissivity of Ti64 are rare and to the best of our knowledge, no literature data is available for the liquid phase of Ti64.

6. Conclusions

We measured the surface tension, viscosity, electrical resistivity, mass density, total hemispheric emissivity, and specific heat capacity of Ti64 alloy by methods using electromagnetic levitation in microgravity. The permanent microgravity conditions on the international space station allow for longer measurements than in PF, which open the door to the precise measurement of, e.g., specific heat capacity and total hemispherical emissivity in the stable and undercooled liquid phase.

The obtained surface tension data exhibit very good agreement with literature data as well as with model predicted values. We have suggested predictive models that can be further developed, to generalize the measured surface tension and viscosity for Ti64 to a broader range of Ti–Al alloys. In this case, the overall agreement with experiment is certainly reasonable, due to successful application of four theoretical models. Concerning the viscosity, the minimum difference (of about 5%) between the new experimental data and calculated values is obtained by applying the MH model, otherwise larger differences were observed.

Electrical resistivity and mass density in the liquid phase are both in agreement with measurements that were performed with different methods; however, we report values having a smaller measurement uncertainty than the present literature. The specific heat capacity obtained in this study aligns with literature data and extends the known data into the undercooled liquid phase. The thermal conductivity in the liquid phase, obtained using the Wiedemann–Franz law, deviates from the values

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derived from diffusivity measurements.[53] This needs further investigation.

However, the uncertainties of the presented approach are smaller than in other methods reported in the literature because the influence of container walls is diminished and the precision is mainly determined by the accuracy of the measurement of a temperature oscillation by a pyrometer.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

electrical resistivity, mass density, specific heat capacity, surface tension, thermal conductivity, total hemispherical emissivity, viscosity

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